@AI Engineer

RAG

Retrieval-Augmented Generation

Interview Question & Answers



DEVENDHIRAN
AI Engineer

@AI Engineer

Overview of RAG Questions

1. **What problem does RAG solve that standalone LLMs cannot?**

RAG enhances LLMs by providing explicit knowledge lookup, allowing for more accurate and reliable answers using real documents instead of memory.

2. **Walk through a basic RAG pipeline end to end.**

A traditional RAG pipeline includes building a knowledge base and processing user queries to retrieve and generate answers with citations.

3. **What roles do the retriever and generator play, and how are they coupled?**

The retriever fetches context for answers, while the generator synthesizes responses, ensuring that the prompt shapes what the generator sees.

@DEVENDHIRAN

@AI Engineer

4. **How does RAG reduce hallucinations compared to pure generation?**

RAG provides evidence for models to quote, anchoring responses to retrieved texts, which shifts the focus from guessing to citing present information.

5. **What types of data sources are commonly used in RAG systems?**

RAG systems utilize various data sources, including internal documents, operational data, engineering content, and structured web data to enhance response accuracy.

6. **What is a vector embedding, and why is it essential for dense retrieval?**

Embeddings numerically represent text, enabling semantic similarity to be captured geometrically, thus improving retrieval efficiency in dense search scenarios.

@DEVENDHIRAN

@AI Engineer

7. **What is chunking, and why does chunk size matter?**

Chunking involves splitting documents into smaller passages for better indexing and retrieval, optimizing the size to maintain contextual relevance and avoid oversaturation.

8. **What is the difference between retrieval and search in RAG contexts?**

Retrieval encompasses broader methods beyond keyword matching, including semantic vector search and metadata filters, thus determining which information is trusted for the model.

9. **What is a vector database, and what problem does it solve?**

Vector databases store embeddings, enabling fast similarity searches, which is critical as data scales, ensuring efficient retrieval without losing indexing capabilities.

@DEVENDHIRAN

@AI Engineer

10. Why is prompt design still critical even when retrieval is involved?

Effective prompt design dictates how retrieved text is utilized, ensuring clarity, structure, and adherence to context, preventing misinterpretations and enhancing output quality.

11. What are common real-world use cases for RAG today?

RAG powers diverse applications including AI search engines, customer support tools, and report drafting systems, demonstrating its versatility in real-world problem-solving.

12. In simple terms, why is RAG preferred over frequent model retraining?

RAG allows for quick and cost-effective updates by refreshing knowledge through document updates rather than the complex and resource-intensive model retraining process.

@DEVENDHIRAN

13. **Compare sparse, dense, and hybrid retrieval methods.**

Sparse methods focus on exact word matches, dense retrieval emphasizes semantic similarity, while hybrid approaches combine both for comprehensive information retrieval strategies.

14. **When would BM25 outperform dense retrieval in a RAG system?**

BM25 excels in scenarios requiring precise token matches, such as part numbers or legal clauses, where semantic interpretation may lead to inaccuracies in retrieval.

15. **How do you decide optimal chunk size and overlap for a given corpus?**

Optimal chunk sizes depend on data structure, with medium chunks for complex documents and smaller ones for FAQs, while adjustments are made based on retrieval performance.

16. What retrieval metrics would you use to measure relevance quality?

Metrics like Recall@k, Precision@k, and MRR help evaluate retrieval effectiveness, ensuring that relevant information is captured and presented accurately to the model.

17. How do you evaluate the final answer quality of a RAG system?

Answer quality assessment involves checking correctness, completeness, faithfulness, citation quality, and helpfulness against established benchmarks and user expectations.

18. What is re-ranking, and where does it fit in the RAG pipeline?

Re-ranking enhances retrieval precision by reordering candidate passages based on relevance after initial retrieval, ensuring that the most pertinent information is prioritized.

@AI Engineer

Follow for more AI Engineer realtime interview questions

DON'T FORGET TO LIKE, COMMENT AND SHARE

